Example 8.6

Calculate the geometric mean of the data given below giving the number of families and the income per head of different classes of people in a village of Kancheepuram District.

Class of people	No. of Families	Income per head in 1990 (Rs)
Landlords	1	1000
Cultivators	50	80
Landless labourers	25	40
Money- lenders	2	750
School teachers	3	100
Shop-keepers	4	150
Carpenters	3	120
Weavers	5	60

Solution:

Calculation of Geometric Mean

Class of people	Income per head in 1990 (₹) X	No of Families f	$\log X$	$f \log X$
Landlords	1000	1	3.0000	3.0000
Cultivators	80	50	1.9031	95.1550
Landless labourers	40	25	1.6021	40.0525
Money- lenders	750	2	2.8751	5.7502
School teachers	100	3	2.0000	6.0000
Shopkeepers	150	4	2.1761	8.7044
Carpenters	120	3	2.0792	6.2376
Weavers	60	5	1.7782	8.8910
		N = 93		Σ=173.7907

Table : 8.4

$$GM = Anti\log\left(\frac{\Sigma f \log X}{N}\right)$$

$$= Anti\log\left(\frac{173.7907}{93}\right)$$

$$= Anti\log(1.8687)$$

GM = 73.91

Example 8.7

Compute the Geometric mean from the data given below:

Marks	0-10	10-20	20-30	30-40	40-50
No. of Students	8	12	18	8	6

Solution:

Marks	m	f	$\log m$	$f \log m$
0-10	5	8	0.6990	5.5920
10-20	15	12	1.1761	14.1132
20-30	25	18	1.3979	25.1622
30-40	35	8	1.5441	12.3528
40-50	45	6	1.6532	9.9192
		N = 52		$\Sigma f \log m$ = 67.1394

Table : 8.5

$$GM = Anti\log\left(\frac{\Sigma f \log m}{N}\right)$$

$$= Anti\log\left(\frac{67.1394}{52}\right)$$

$$= Anti\log(1.2911)$$

GM = 19.54

Specific uses of Geometric mean

The most useful application of geometric mean is to average the rate of changes. For example, from 2006 to 2008 prices increased by 5%, 10% and 18% respectively. The average annual increase is not 11% $\left(\frac{5+10+18}{3} = 11\right)$ as given by the arithmetic average but 10.9% as obtained by the geometric mean. This average is also useful in measuring the growth of population, because population increases in geometric progression.

Example 8.8

Compared to the previous year the overhead expenses went up by 32% in 1995, they

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